JEREMY MCCORMICK

Data Engineer

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About

Data Engineer with experience building and productionizing cloud-native data pipelines across startup and enterprise environments. Skilled in designing scalable ETL/ELT workflows, serverless lakehouse architectures, and analytics-ready data models using AWS, DBT, Apache Iceberg, and modern CI/CD practices. Experienced integrating diverse data sources including APIs, IoT telemetry, geospatial, and vendor systems to support operational analytics, regulatory reporting, and business decision-making.

Skills

- **Data Engineering:** Python, SQL, DBT, Apache Airflow, Apache Spark, Apache Iceberg
- **Cloud Platforms:** AWS (S3, Lambda, Athena, Glue, EC2), Terraform, Databricks (Delta Lake), GCP (BigQuery, Spanner)
- **Data Modeling:** Dimensional Modeling, Medallion Architecture, Data Lakehouse Architecture
- **Databases:** PostgreSQL, Snowflake, BigQuery, Neo4j, Azure SQL Server
- **Orchestration & CI/CD:** Airflow, GitHub Actions
- **Business Intelligence:** Tableau, Power BI, Looker, Streamlit, Dash
- **Other:** REST APIs, GeoPandas

Work Experience

DATA ENGINEER (CONTRACT) – Townn – Remote — May 2025 – Present

- Designed near-real-time data ingestion workflows and **DBT** transformation pipelines for operational analytics, with automated **CI/CD** deployments to **GCP via GitHub Actions** and **Git**
- Implemented automated **DBT** unit testing workflows using **Python** Faker-generated synthetic application data and **DBT** fixtures in G**itHub Actions CI/CD**, enabling test-driven validation and automated merge approvals
- Collaborated with stakeholders to **define key reporting needs** and built initial dashboards in **Looker**, providing the first data-driven insights into core business operations.
- Operated within an **Agile development environment**, managing and executing data engineering work tickets in **Jira** across sprints in a fast-paced startup setting.

DATA ANALYST – Atmos Energy – Jackson, MS — October 2023 – Present

- Built and **productionized** an end-to-end **ETL** pipeline (**Python, SQLite, Snowflake**) to automate the Damage Per Thousand metric; ingested vendor CSV reports via email parsing, staged data locally, and joined against warehouse damage data to generate operational KPIs that improved visibility into pipeline incident rates and decision-making
- **Engineered a multi-source ETL pipeline (REST APIs + IoT sensor data)** to support a corrosion risk model, enabling proactive identification of high-risk pipeline segments
- **Developed and maintained data models in Azure SQL Server**, supporting internal engineering workflows and project lifecycle tracking
- **Designed and implemented a Neo4j graph database** to model pipeline asset networks, improving traversal of infrastructure relationships and operational visibility
- **Performed geospatial data processing (GeoPandas, ESRI API)** to spatially relate pipeline assets, supporting graph modeling and field operations
- Developed operational metrics that contributed to a **56-day reduction in leak repair times**, improving efficiency and regulatory performance
- Identified **$80K in unaccounted capital**, improving financial accuracy through data analysis

DATA ANALYTICS APPRENTICE - Nashville Software School — May 2023 - November 2023

- Completed intensive full-time program; built ETL pipelines and performed data transformations using **Python (Pandas, NumPy)**, queried relational databases with **PostgreSQL**, applied geospatial analysis with **GeoPandas**, ingested data via **REST APIs and web scraping**, and managed source control with **Git/GitHub**

OWNER/EDUCATOR – Spot's Drum Hub – Nashville, TN December 2016 - October 2023

- Leveraged **Google Analytics** and **MailChimp** campaign data to track student engagement and optimize outreach, growing roster to **100 weekly clients**.

Projects

AWS FMCSA Lakehouse **–** Personal Project May 2026

- Built and provisioned a fully serverless AWS lakehouse architecture using **Terraform**, deploying **Lambda** ingestion, **IAM policies**, **EventBridge** scheduling, **S3 storage**, and **Glue Catalog** resources entirely through infrastructure-as-code
- Engineered an automated monthly ingestion pipeline using **AWS Lambda** and **Python** to pull FMCSA carrier census data from the Socrata REST API into Amazon S3 as raw JSON bronze-layer storage
- Modeled a **medallion architecture** using **Athena, DBT**, and **Apache Iceberg** to transform raw API data into partitioned Parquet analytics tables optimized for downstream BI and reporting workloads
- Developed **Tableau** analytical marts aggregating active carrier counts, power units, and driver metrics by state, business organization type, and reporting year

World Flight Tracker **–** Personal Project May 2026

- Engineered an end-to-end **medallion architecture pipeline** ingesting near-real-time global flight telemetry from the OpenSky Network REST API into PostgreSQL as a raw JSON bronze layer, processing ~6.5M flight telemetry records daily
- **Flattened and normalized JSON into structured data marts** (silver layer) in PostgreSQL, then exported Parquet files every 5 minutes to Amazon S3 as the gold layer to support near-real-time analytical workloads
- Integrated **Databricks Delta Lake** to consume S3 Parquet data into ACID-compliant analytical views, implementing modern lakehouse architecture patterns using **Delta Lake**, columnar storage, and incremental refresh workflows

California Drought & Weather Correlation Tool - Personal Project November 2025

- Engineered a scalable AWS **Data Lakehouse** using **Apache Iceberg** to manage schema evolution and time-travel capabilities for multi-source environmental datasets
- Implemented **a Medallion Architecture** (Bronze, Silver, Gold) to orchestrate data refinement, utilizing **Apache Iceberg** for schema enforcement and ACID transactions across the data lakehouse
- Architected an automated ETL pipeline using **PySpark**, **Airflow**, and **DBT** to ingest and normalize **GeoJSON** (ca.gov) and REST API (Weather/Drought) data into a unified **S3**-backed storage layer.
- Optimized query performance by implementing columnar storage and partitioning strategies, reducing data retrieval latency for downstream BI tools

Education

BACHELORS IN SCIENCE – Liberty University – Lynchburg, VA 2024
Majors: Business Administration & Data Analysis: Data Analytics

DATA ANALYTICS CERTIFICATE – Nashville Software School – Nashville, TN 2023
Majors: Data Analytics

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